

Lucas Wu

Electrical Engineering | Available Jan–Apr 2027

Waterloo, ON / Calgary, AB | lucaswu2018@gmail.com

EDUCATION

University of Waterloo

Sep 2026–Apr 2031 (expected)

Bachelor of Applied Science candidate, Electrical Engineering (Honours, Co-op) | First year

Current coursework: Fundamentals of Programming (C++), Project Studio, Linear Algebra, Calculus

Western Canada High School | Graduated Jun 2026 | Admission average: 96%

TECHNICAL SKILLS

Digital design: Verilog, SPI, clock-domain crossing (CDC), cocotb, Tiny Tapeout

Robotics: VEX V5 competition robot design, assembly, and programming

Programming and tools: C++, Python, Java, JavaScript, HTML/CSS; Git, WSL; NumPy, OpenCV, MediaPipe, Flask

Engineering tools: KiCad (tutorial-based water sensor design), MATLAB

TECHNICAL PROJECTS

UWASIC | RTL Design Member, Ray-Tracing ASIC | [Onboarding code](#)

2026–Present

- Currently designing functional-unit RTL for UWASIC's ray-tracing ASIC, targeting a lint-clean RTL-to-GDS flow and passing functional tests in collaboration with the verification team.
- Completed onboarding: built a Verilog SPI Mode 0 peripheral with 5 registers, 16 outputs, and two-stage CDC.
- Wrote Python/cocotb tests for 3 kHz PWM, 0%/50%/100% duty cycles, and output-enable behavior.
- Configured Tiny Tapeout and GitHub Actions to run simulation tests and generate GDS layout files.

VEX Robotics | Competition Robot

2020–2023

- Designed and built a VEX V5 robot and programmed its autonomous routines, helping the team place second in Alberta Provincial Robot Skills.
- Designed and built the drivetrain, disc intake, and flywheel launcher; iteratively tuned flywheel speed and intake spacing to improve shooting consistency and reduce jams.
- Qualified for and competed at the VEX Robotics World Championship in Dallas in 2023.

SignEd | Hack the Bias — Beginner Track Winner | [Project code](#)

2026

- Developed an ASL learning website as the team's sole programmer using Python, Flask, JavaScript, and HTML/CSS.
- Recognized ASL poses with MediaPipe/OpenCV and distance-weighted KNN; normalized 21 3D hand landmarks by wrist position and hand size.

EXPERIENCE AND LEADERSHIP

Coding Instructor | Code Ninjas

May–Aug 2026

- Taught programming through game-development projects; provided individual debugging support and adapted lessons to student progress.

Co-Founder and Web Designer | Calgary Ace Learning Center

Mar 2024–Present

- Co-founded a student-run nonprofit; contributed 400+ volunteer hours tutoring and organizing, and led fundraising that raised \$5,000+ for the Alberta Children's Hospital Foundation.
- Built the organization's website using WordPress.

President | Western Canada High School Computer Science Club

Sep 2024–Jun 2026

- Led weekly sessions for a 20+ member club; created teaching materials and mentored students in competitive programming and problem solving.

ADDITIONAL EXPERIENCE AND ACHIEVEMENTS

CCC Distinction: Junior (2022), Senior (2023) | Canadian Senior Mathematics Contest Distinction (2022).

VP of IT, RECAPD / Junior Achievement (Oct 2024–Apr 2025): built and maintained a Shopify website.

Coding Teaching Assistant, ABAITC (Jul–Aug 2024): supported lessons and helped students debug projects.

Level 2 Hockey Referee (Oct 2023–Present) | 11 seasons of competitive hockey | English and Mandarin